

Exploratory Data Analysis - Phase 3

In [1]: *# Load Processed Data*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

train_df = pd.read_csv("data/processed/train_2010_2022.csv")
test_df = pd.read_csv("data/processed/test_2023_2025.csv")
predict_2026_df = pd.read_csv("data/processed/predict_2026.csv")

print("Train:", train_df.shape)
print("Test:", test_df.shape)
print("Predict 2026:", predict_2026_df.shape)
```

Train: (120, 17)

Test: (34, 17)

Predict 2026: (10, 17)

In [2]: *# Combine Train and Test for EDA*

```
eda_df = pd.concat([train_df, test_df], axis=0)

print("EDA Dataset:", eda_df.shape)
eda_df.head()
```

EDA Dataset: (154, 17)

Out[2]:

	player_id	player_name	season	position	nfl_team	draft_round	draft_pick_overall	age_on_draft_day	height_inches
0	qb_2010_bradford	Sam Bradford	2010	QB	Rams	1	1	22	76
1	rb_2010_spiller	C.J. Spiller	2010	RB	Bills	1	9	23	71
2	rb_2010_mathews	Ryan Mathews	2010	RB	Chargers	1	12	22	72
3	te_2010_gresham	Jermaine Gresham	2010	TE	Bengals	1	21	22	77
4	wr_2010_thomas	Demaryius Thomas	2010	WR	Broncos	1	22	22	75

In [3]:

Check Seasons

eda_df["season"].value_counts().sort_index()

Out[3]:

```

season
2010      8
2011      8
2012     11
2013      5
2014      9
2015     10
2016      8
2017     11
2018     11
2019      8
2020     11
2021     13
2022      7
2023     10
2024     14
2025     10
Name: count, dtype: int64

```

In [4]:

Check Positions

```
eda_df["position"].value_counts()
```

```
Out[4]: position
WR      66
QB      51
RB      23
TE      14
Name: count, dtype: int64
```

```
In [5]: # Summary of Numeric Columns
```

```
eda_df.describe()
```

```
Out[5]:
```

	season	draft_round	draft_pick_overall	age_on_draft_day	height_inches	weight_lbs	college_games_played	career_
count	154.000000	154.0	154.000000	154.000000	154.000000	154.000000	154.000000	154.00
mean	2017.980519	1.0	14.376623	21.870130	73.707792	215.441558	37.785714	4521.60
std	4.592507	0.0	9.803346	0.994766	2.556359	19.328595	8.600018	3279.45
min	2010.000000	1.0	1.000000	20.000000	67.000000	165.000000	12.000000	778.00
25%	2014.000000	1.0	6.000000	21.000000	72.000000	204.250000	32.000000	2316.25
50%	2018.000000	1.0	12.000000	22.000000	74.000000	217.000000	38.000000	3156.00
75%	2022.000000	1.0	23.000000	22.000000	76.000000	226.750000	41.750000	6201.00
max	2025.000000	1.0	32.000000	28.000000	79.000000	260.000000	61.000000	15351.00

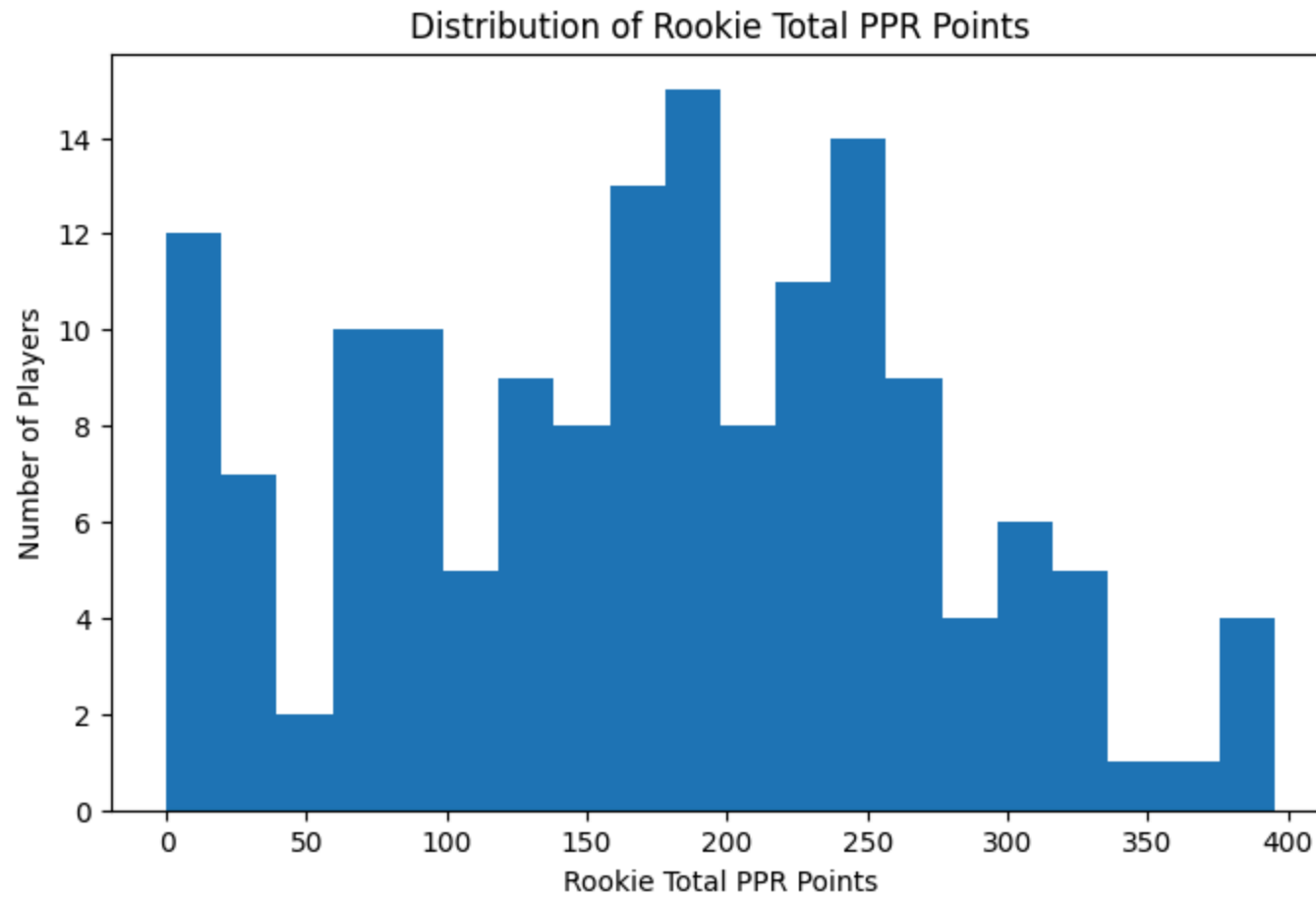


```
In [6]: # Check Target Distribution
```

```
eda_df["rookie_total_ppr_points"].describe()
```

```
Out[6]: count    154.000000  
        mean     173.757792  
        std       97.296186  
        min        0.000000  
        25%       94.975000  
        50%      179.800000  
        75%      242.550000  
        max      395.100000  
        Name: rookie_total_ppr_points, dtype: float64
```

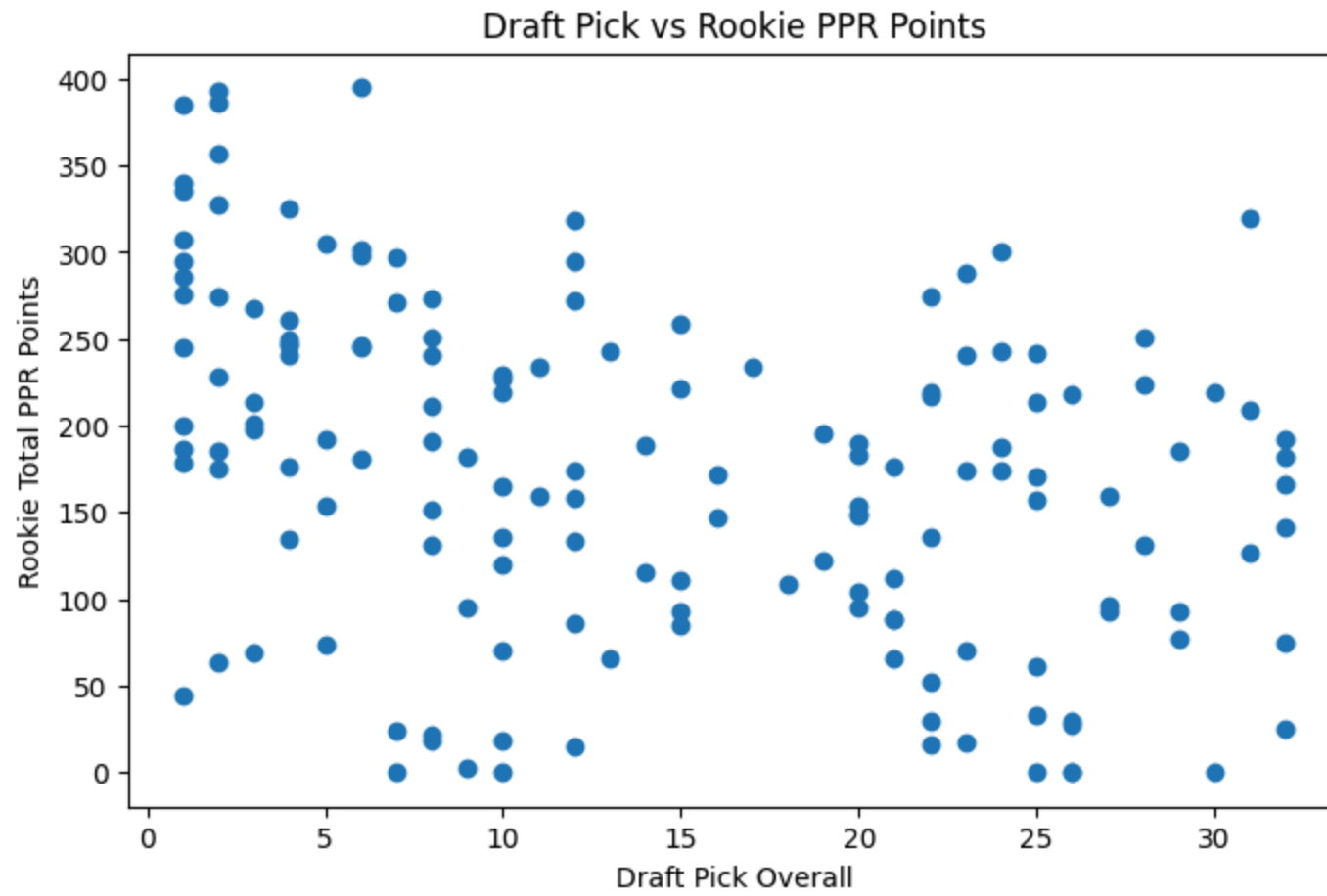
```
In [7]: # Plot Target Distribution  
  
plt.figure(figsize=(8, 5))  
plt.hist(eda_df["rookie_total_ppr_points"].dropna(), bins=20)  
plt.xlabel("Rookie Total PPR Points")  
plt.ylabel("Number of Players")  
plt.title("Distribution of Rookie Total PPR Points")  
plt.show()
```



In [8]: *# Draft Pick vs Fantasy Points*

```
plt.figure(figsize=(8, 5))
plt.scatter(
    eda_df["draft_pick_overall"],
    eda_df["rookie_total_ppr_points"]
)

plt.xlabel("Draft Pick Overall")
plt.ylabel("Rookie Total PPR Points")
plt.title("Draft Pick vs Rookie PPR Points")
plt.show()
```



```
In [9]: # Average Fantasy Points by Position
```

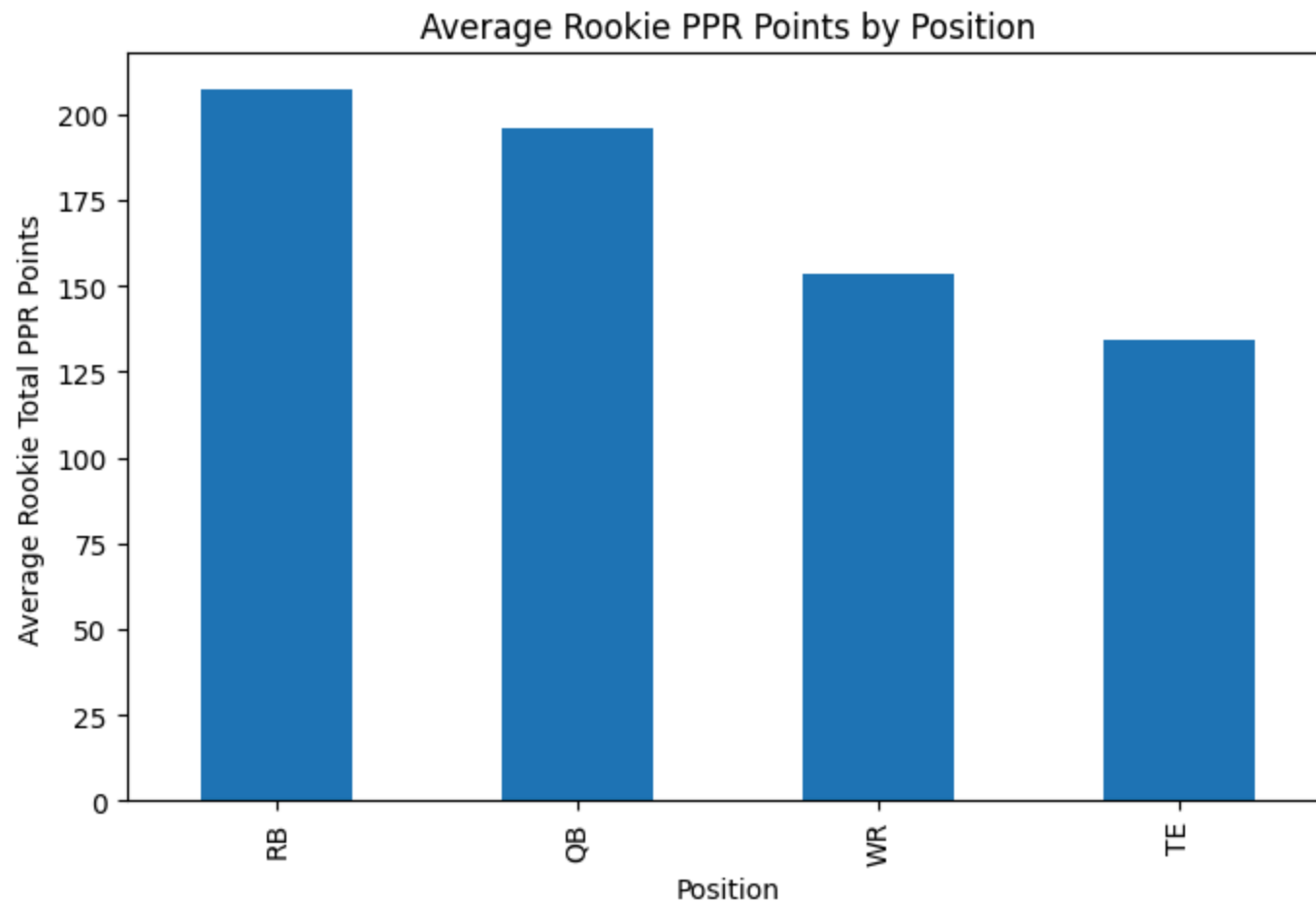
```
eda_df.groupby("position")["rookie_total_ppr_points"].mean().sort_values(ascending=False)
```

```
Out[9]: position
RB      207.386957
QB      195.664706
WR      153.484848
TE      134.278571
Name: rookie_total_ppr_points, dtype: float64
```

```
In [10]: # Plot - Average Fantasy Points by Position
```

```
position_avg = eda_df.groupby("position")["rookie_total_ppr_points"].mean().sort_values(ascending=False)

plt.figure(figsize=(8, 5))
position_avg.plot(kind="bar")
plt.xlabel("Position")
plt.ylabel("Average Rookie Total PPR Points")
plt.title("Average Rookie PPR Points by Position")
plt.show()
```



In [11]: *# Fantasy Relevant Count*

```
eda_df["fantasy_relevant_binary"].value_counts()
```

```
Out[11]: fantasy_relevant_binary
0.0    91
1.0    63
Name: count, dtype: int64
```

```
In [12]: # Fantasy Relevant Count (Normalized)

eda_df["fantasy_relevant_binary"].value_counts(normalize=True)
```

```
Out[12]: fantasy_relevant_binary
0.0    0.590909
1.0    0.409091
Name: proportion, dtype: float64
```

```
In [13]: # Check for duplicate column names
eda_df = eda_df.reset_index(drop=True)
```

```
In [ ]: # Chart - Fantasy Relevance by Position

relevance_by_position = pd.crosstab(
    eda_df["position"],
    eda_df["fantasy_relevant_binary"],
    normalize="index"
) * 100

relevance_by_position.round(1)
```

```
Out[ ]: fantasy_relevant_binary    0.0    1.0
```

	position	
	QB	RB
0.0	84.3	34.8
1.0	15.7	65.2
	TE	WR
0.0	42.9	51.5
1.0	57.1	48.5